GE MRI

7 K-SPACE ARTIFACTS – CAUSES & APPEARANCES

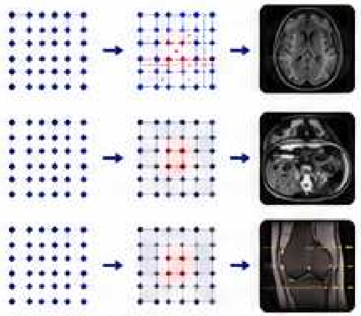
Most artifacts are caused by errors in k-space sampling.

THE BIG IDEA
Artifacts are predictable. Understanding the source in k-space helps you prevent, reduce, or recognize them quickly.

THE BASICS

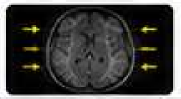
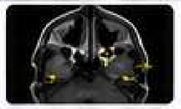
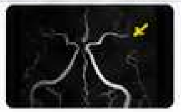
1. MRI assumes the signal is stable while k-space is sampled.
2. Patient motion, system instability, or incorrect sampling violate this assumption.
3. These errors show up as artifacts in the image.
4. The appearance of an artifact gives clues to its cause.
5. Fix the cause at the source for the best results.

WHERE ARTIFACTS COME FROM (IN K-SPACE)

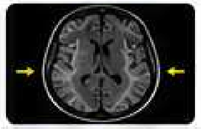
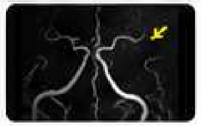


Errors in k-space → errors in image.

COMMON K-SPACE ARTIFACTS

ARTIFACT	CAUSE (K-SPACE ERROR)	APPEARANCE (WHAT YOU SEE)	TYPICAL LOCATIONS / PATTERNS	EXAMPLE IMAGE
Motion (Ghosting)	Phase inconsistencies between repetitions (line-to-line phase errors)	Repeating ghosts (usually in phase encoding direction)	Along phase encoding (ky) direction, equally spaced	
Wrap-around (Aliasing)	Under-sampling: not enough phase encoding lines (Ny < object size)	Anatomy appears on opposite side of image	Opposite side (fold-over) along phase encoding direction	
Truncation (Ringing)	Too little data in k-space (sharp cutoff) High contrast edges	Dark and bright bands (ringing) near sharp edges	Bands parallel to phase encoding direction near high contrast boundaries	
Chemical Shift	Different resonant frequencies of fat and water	Bright/dark edge misregistration	At fat-water boundaries (phase encoding direction)	
Susceptibility	Local B0 field variations (magnetic susceptibility differences)	Signal loss, distortion, blurring, pile-up	Near air-tissue or metal (e.g., sinuses, ear, implants)	
Flow (Phase Dispersion)	Moving spins change phase during readout	Signal loss or flow voids	In vessels or CSF (flowing blood/CSF)	

VISUAL GUIDE – ARTIFACT PATTERNS

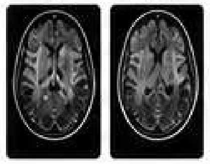
ARTIFACT	K-SPACE ERROR (CONCEPT)	DIRECTION / PATTERN	IMAGE APPEARANCE (EXAMPLE)
Motion (Ghosting)	Phase inconsistency between lines	Ghosts along phase encoding (ky) direction	
Wrap-around (Aliasing)	Not enough phase encoding lines	Fold-over on opposite side (ky) direction	
Truncation (Ringing)	Sharp cutoff in k-space	Ringing (bands) parallel to ky direction	
Chemical Shift	Fat and water frequency difference	Misregistration along ky direction	
Susceptibility	Local field inhomogen-eity. (distortion)	Geometric distortion & signal loss	
Flow (Phase Dispersion)	Moving spins lose phase coherence	Signal loss in flowing areas (no direction)	

HOW TO REDUCE OR ELIMINATE ARTIFACTS

ARTIFACT	WHAT HELPS	TECHNIQUES / TIPS
Motion (Ghosting)	Reduce motion	Patient coaching, faster scans, PROPELLER/BLADE, gating, navigator echoes, sedation
Wrap-around (Aliasing)	Increase phase FOV	Increase FOV, increase phase matrix, use oversampling, apply anti-alias filters
Truncation (Ringing)	Add more data / Smooth edges	Increase matrix, increase FOV, use apodization (filters), avoid sharp truncation
Chemical Shift	Narrow bandwidth	Use lower BW, fat sat, Dixon, view order in frequency direction
Susceptibility	Minimize field inhomogeneity	Shorter TE, higher BW, shimming, SEMAC/MAVRIC, avoid metal when possible
Flow (Phase Dispersion)	Compensate for flow	Flow compensation, gradient moment nulling, faster readout, black-blood techniques

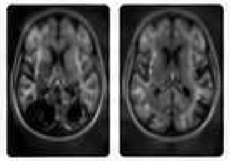
REAL-WORLD EXAMPLES (RECOGNIZE & FIX)

Motion (Ghosting)



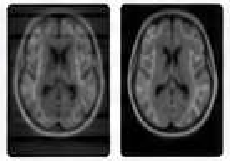
TIP: Use faster scan or motion correction.

Wrap-around (Aliasing)



TIP: Increase phase FOV or matrix.

Truncation (Ringing)

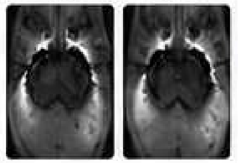


TIP: Increase matrix or use filtering.

Chemical Shift

TIP: Use fat sat or narrow bandwidth.

Susceptibility



TIP: Shorter TE, higher BW, or advanced shimming.

Flow (Phase Dispersion)

TIP: Use flow comp or black-blood techniques.

KEY TAKEAWAYS

- Artifacts are signs of k-space problems.
- The pattern tells you the cause.
- Fix the cause at the source.
- Prevention > correction.
- Better setup = fewer artifacts = better diagnosis.

BOTTOM LINE

Motion → Ghosting
Phase errors between repetitions

Under-sampling → Alias
Not enough lines in k-space

Truncation → Ringing
Cutting k-space too short

Chemical Shift → Misregistration
Fat and water frequency difference

Susceptibility → Distortion
B0 inhomogeneity near air or metal

Flow → Signal loss
Moving spins lose phase coherence